Qi Wei Mi Ma: Future Quality of Life
Odor Code Report

Theoretical and Practical Comprehensive Research Report - Theory & Practice Integration

Executive Summary: Humans spend 90% time indoors, indoor pollution 2-5x outdoor. UEA 2026 shows smell loss impact equals diabetes/stroke/Parkinson. This report builds Odor-Health-QoL closed loop across olfactory neuroscience, environmental health, AI e-nose, smart toilet and canteen health, and proposes 5-year path for AERISENS Pro.

Chapter 1: Why Odor is the Underlying Code of QoL

Humans detect H2S at ppb level but 100ppm paralyzes olfactory nerve in 3-15min, 150ppm complete paralysis - paradox of more dangerous less smellable. UEA 2026 review of 40+ studies shows QoL scores of smell-loss patients equal diabetes/stroke. Core suffering: food becomes functional, social rituals collapse, safety anxiety for gas/smoke, emotional numbness. Pollutants enter brain via thin nasal mucosa damaging limbic system responsible for emotion/memory.



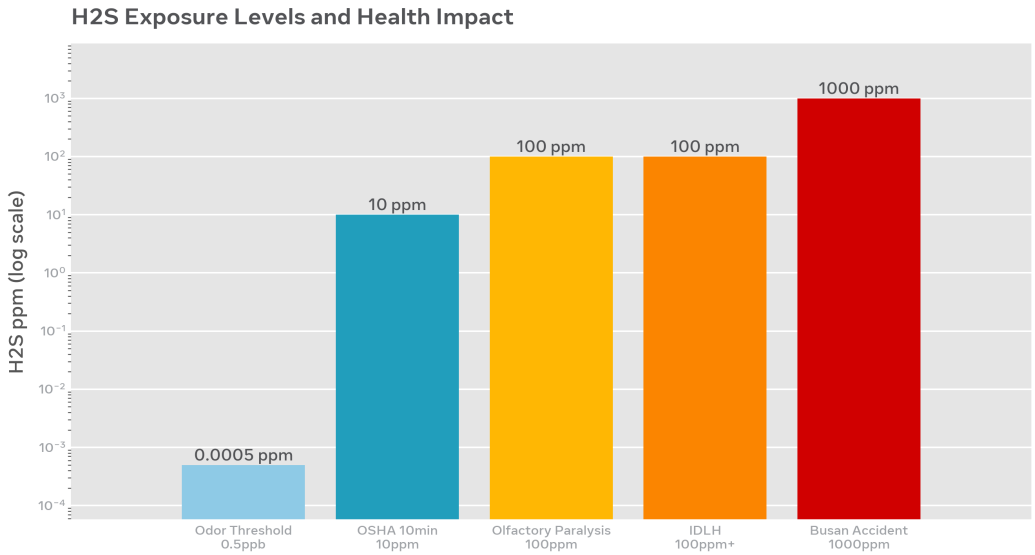
Chapter 2: Evolution from Human Nose to AI E-Nose

Gen1 MOS (Figaro TGS, Winsen MQ) - cross sensitivity; Gen2 Electrochemical+NDIR (Honeywell, MSA) - $950 industrial; Gen3 AI E-Nose (Renesas ZMOD4410, DGIST LIG) - combinatorial coding + neural network. Renesas ZMOD4410 with e-AI distinguishes sulfur vs ethanol odors for bathroom control, IP67 version released. DGIST 2025 breakthrough: single-step laser fabrication of porous LIG array, >95% accuracy for 9 fragrances, 30k bends, published in ACS Nano 2025. Neuromorphic olfactory chips nearing human level per Nature Reviews Electrical Engineering 2025.

Chapter 3: Field Practice - Toilet and Canteen

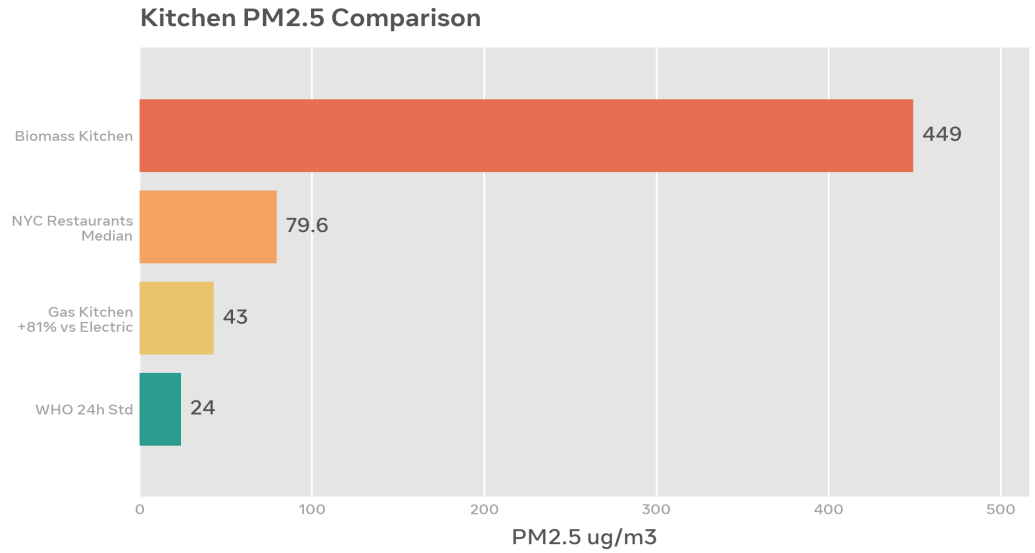
3.1 Smart Toilet - From Toilet Revolution to Digital Nose

Policy: Central fiscal 7B CNY/year for toilet revolution, Shanghai renovated 1200 toilets in 2024. Risk: Busan case 1000ppm H2S 60x over limit causing brain damage death. OSHA: 10ppm 10min acceptable, 100+ppm IDLH. China practice: Jinan Changqing platform - IoT sensors for NH3/H2S real-time, auto alarm, task dispatch, 99% online rate. Guiyang Arrow Street smart toilet + auto fragrance. Chengdu 5G toilet with laser sensor, mmWave human detection, environmental dashboard.



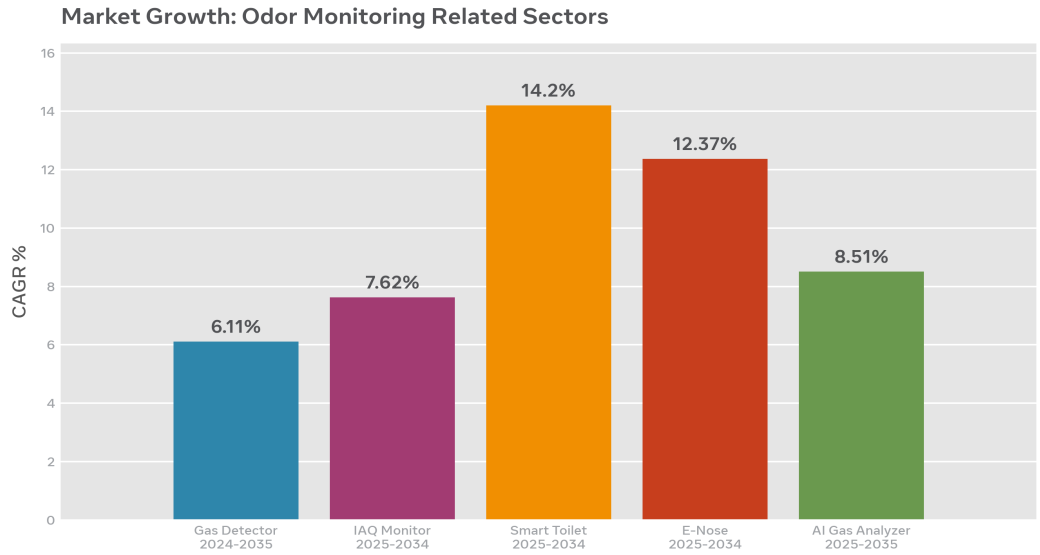
3.2 Canteen - Invisible Lung Cancer Risk

Gas Chinese kitchen PM2.5/TVOC +81%/78% vs electric, NYC 18 restaurants median 79.6 ug/m3 vs WHO 24. Canteen HOVCs 53-357 ug/m3. Epidemiology confirms cooking fume linked to non-smoking lung cancer and kidney microalbuminuria. Standard GB 18483-2001 2.0mg/m3 to be tightened to 1.0. Behavior: turning on hood 20s before cooking better controls spread - real-time reminder more effective than post purification.



Chapter 4: Industry Economics

Market: Gas detector $13.27B->25.49B CAGR6.11%; IAQ $5.57B->10.8B 7.62%; Smart toilet $2.72B->8.54B 14.2%; E-Nose $157M->449M 12.37%; AI analyzer $2.71B->6.12B 8.51%; SMO sensor $1.5B->3.2B 9.2%.



China supply: Hanwei 2024 2.228B CNY, sensor +20.98% in 2025H1; Cubic 2021-2024 547M->873M +26.23%, investing 600M base. Huaqiangbei SEG provides chips/MCU/sensors, 3-hour sampling, 70% global gadgets origin. From chip counter to Dapeng industrial park 5-year journey.

Business model: Hardware+Subscription - Atmocube monitors CO2/VOC/PM1/PM2.5/PM10/temp/humidity/noise with cloud subscription; IAQ as a Service real-time + AI ventilation + compliance report. AERISENS BOM 383 CNY (1k units), price 1299 CNY / $199, 66% margin, consumable 8 CNY/month, LTV 3yr 1587 CNY.

Chapter 5: Five Future Trends of Odor Civilization

1. From detection to intervention: odor alarm -> task dispatch -> rectification closed loop, odor monitoring becomes city governance nerve ending

2. From single point to scenario: toilet needs H2S+NH3+humidity to judge water seal dry, canteen needs PM2.5+TVOC+CO+temp/humidity for ventilation

3. From industrial to consumer: Renesas e-AI makes consumer sensor industrial-grade consistent, $100-200 price structural gap

4. From smell to multi-sensory: Guiyang toilet auto fragrance on threshold, future Detection-Purification-Fragrance-Feedback loop

5. From health to happiness: smell loss equals diabetes, odor management will be part of WHO healthy building standard

Chapter 6: Conclusion

Theoretical contribution: Odor Code model proves odor is not comfort issue but underlying variable via Chemical Exposure - Olfactory Damage - Psychological Load - QoL chain, economic value should be measured in DALY not hardware cost. Practical path: Short 2026 Huaqiangbei ZMOD4410+electrochemical H2S+PM2.5 triple-in-one 1299 CNY, bind Changqing platform as B-end demo; Mid 2027-2028 introduce DGIST laser graphene array 95% recognition for 9 odors, distinguish H2S vs ethanol false alarm, provide root cause diagnosis; Long 2029+ build odor data network as third-party standard for canteen compliance and toilet civilization, subscription >50%. One sentence: When 70% gadgets from Huaqiangbei, when 1000ppm H2S lethal, when smell loss equals stroke, odor monitoring is not optional air freshener but essential infrastructure for future QoL.

References

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Tech: Renesas IoT M2M Council 2020, DGIST ACS Nano 2025, Nature Reviews Electrical Engineering 2025, EDN IP67
Cases: Jinan Changqing 2026, Guiyang 2025, Chengdu 2026, Korea ETtoday 1000ppm, NYC PM2.5 79.6ug/m3
Supply: SimooSourcing Huaqiangbei, Cubic 2025, Hanwei 2025